

Constant propagation optimization replaces variables that hold set values with their constants throughout the code to simplify and improve performance. For instance, consider this example:

```
int i = 10;  
int p = i * 2;  
int result = p + 5;
```

After constant propagation, the compiler sees that **i** is always 10 and **p** is always 20, so it simplifies the code to:

```
int result = 25;
```

In compiler pseudocode, this process might look like:

```
for each assignment s:  
  if variable v is assigned a constant c:  
    record v -> c  
  if expression e uses v and v -> c exists:  
    replace v with c in e
```

This ensures that redundant computations are removed and all constant values are directly substituted before code generation.